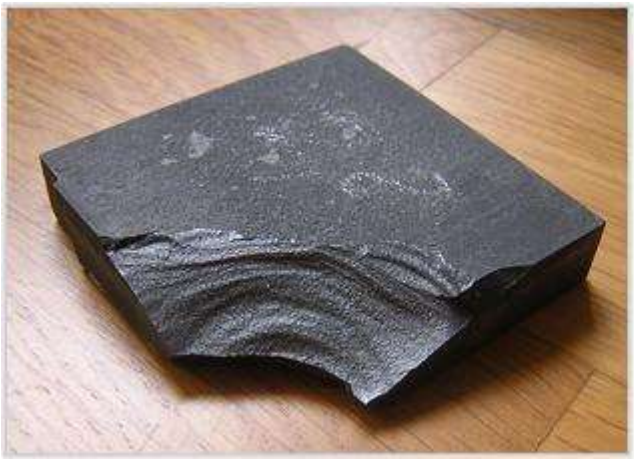
## Description

### Image





### Caption

1. A piece of Boron carbide © Preslav at Wikimedia Commons (CC BY 3.0) 2. High Speed Body Armor © Evike Inc

### The material

Boron carbide (B4C) is nearly as hard as diamond and vastly less expensive (though still not cheap). It's very low density and high hardness make it attractive for the outer layer of bulletproof body armor, as nozzles for sandblasting and as an abrasive.

### Compositional summary

B4C

## General properties

| | | | | | |
|---|---|---|---|---|---|
| Density | | 2.49e3 | - | 2.55e3 | kg/m^3 |
| Price | * | 45.4 | - | 67.3 | GBP/kg |
| Date first used | | 1930 | | | |

## Mechanical properties

| | | | | | |
|---|---|---|---|---|---|
| Young's modulus | | 362 | - | 380 | GPa |
| Shear modulus | | 180 | - | 190 | GPa |
| Bulk modulus | | 218 | - | 230 | GPa |
| Poisson's ratio | | 0.18 | - | 0.2 | |
| Yield strength (elastic limit) | * | 261 | - | 289 | MPa |
| Tensile strength | * | 261 | - | 289 | MPa |
| Compressive strength | * | 2.61e3 | - | 2.89e3 | MPa |
| Elongation | * | 0.07 | - | 0.08 | % strain |
| Hardness - Vickers | | 3.99e3 | - | 4.41e3 | HV |
| Fatigue strength at 10^7 cycles | * | 223 | - | 260 | MPa |

| | | | | | |
|---|---|---|---|---|---|
| Fracture toughness | | 2.8 | - | 3.4 | MPa.m^0.5 |
| Mechanical loss coefficient (tan delta) | * | 1e-5 | - | 3e-5 | |

## Thermal properties

| | | | | | |
|---|---|---|---|---|---|
| Melting point | | 2.4e3 | - | 2.51e3 | °C |
| Maximum service temperature | * | 1.73e3 | - | 1.82e3 | °C |
| Minimum service temperature | | -273 | | | °C |
| Thermal conductor or insulator? | | Poor conductor | | | |
| Thermal conductivity | | 16.7 | - | 17.9 | W/m.°C |
| Specific heat capacity | | 840 | - | 913 | J/kg.°C |
| Thermal expansion coefficient | | 4.41 | - | 4.59 | µstrain/°C |

## Electrical properties

| | | | | | |
|---|---|---|---|---|---|
| Electrical conductor or insulator? | | Poor conductor | | | |
| Electrical resistivity | | 9.48e4 | - | 9.48e5 | µohm.cm |
| Dielectric constant (relative permittivity) | | 4.8 | - | 5.2 | |
| Dissipation factor (dielectric loss tangent) | * | 0.001 | - | 0.01 | |
| Dielectric strength (dielectric breakdown) | * | 5 | - | 10 | MV/m |

## Optical properties

| | | | | | |
|---|---|---|---|---|---|
| Transparency | | Opaque | | | |

## Critical Materials Risk

| | | | | | |
|---|---|---|---|---|---|
| High critical material risk? | | No | | | |

## Processability

| | | | | | |
|---|---|---|---|---|---|
| Moldability | | 2 | - | 3 | |
| Machinability | | 1 | - | 2 | |

## Durability: water and aqueous solutions

| | | | | | |
|---|---|---|---|---|---|
| Water (fresh) | | Excellent | | | |
| Water (salt) | | Excellent | | | |
| Soils, acidic (peat) | | Excellent | | | |
| Soils, alkaline (clay) | | Excellent | | | |
| Wine | | Excellent | | | |

## Durability: acids

| | | | | | |
|---|---|---|---|---|---|
| Acetic acid (10%) | | Excellent | | | |
| Acetic acid (glacial) | | Excellent | | | |
| Citric acid (10%) | | Excellent | | | |

| | |
|---|---|
| Hydrochloric acid (10%) | Excellent |
| Hydrochloric acid (36%) | Excellent |
| Hydrofluoric acid (40%) | Acceptable |
| Nitric acid (10%) | Excellent |
| Nitric acid (70%) | Excellent |
| Phosphoric acid (10%) | Excellent |
| Phosphoric acid (85%) | Excellent |
| Sulfuric acid (10%) | Excellent |
| Sulfuric acid (70%) | Excellent |

## Durability: alkalis

| | |
|---|---|
| Sodium hydroxide (10%) | Excellent |
| Sodium hydroxide (60%) | Excellent |

## Durability: fuels, oils and solvents

| | |
|---|---|
| Amyl acetate | Excellent |
| Benzene | Excellent |
| Carbon tetrachloride | Excellent |
| Chloroform | Excellent |
| Crude oil | Excellent |
| Diesel oil | Excellent |
| Lubricating oil | Excellent |
| Paraffin oil (kerosene) | Excellent |
| Petrol (gasoline) | Excellent |
| Silicone fluids | Excellent |
| Toluene | Excellent |
| Turpentine | Excellent |
| Vegetable oils (general) | Excellent |
| White spirit | Excellent |

## Durability: alcohols, aldehydes, ketones

| | |
|---|---|
| Acetaldehyde | Excellent |
| Acetone | Excellent |
| Ethyl alcohol (ethanol) | Excellent |
| Ethylene glycol | Excellent |
| Formaldehyde (40%) | Excellent |
| Glycerol | Excellent |
| Methyl alcohol (methanol) | Excellent |

## Durability: halogens and gases

| | |
|---|---|
| Chlorine gas (dry) | Excellent |
| Fluorine (gas) | Limited use |
| O2 (oxygen gas) | Acceptable |
| Sulfur dioxide (gas) | Excellent |

## Durability: built environments

| | |
|---|---|
| Industrial atmosphere | Excellent |
| Rural atmosphere | Excellent |
| Marine atmosphere | Excellent |
| UV radiation (sunlight) | Excellent |

## Durability: flammability

| | |
|---|---|
| Flammability | Non-flammable |

## Durability: thermal environments

| | |
|---|---|
| Tolerance to cryogenic temperatures | Excellent |
| Tolerance up to 150 C (302 F) | Excellent |
| Tolerance up to 250 C (482 F) | Excellent |
| Tolerance up to 450 C (842 F) | Excellent |
| Tolerance up to 850 C (1562 F) | Excellent |
| Tolerance above 850 C (1562 F) | Excellent |

## Primary material production: energy, climate change and water

| | | | | |
|---|---|---|---|---|
| Climate change (CO2-eq) (virgin grade production) | * 8.23 | - | 9.1 | kg/kg |
| Embodied energy (virgin grade production) | * 153 | - | 169 | MJ/kg |
| Water usage | * 87.5 | - | 262 | l/kg |

## Material processing: energy

| | | | | |
|---|---|---|---|---|
| Machining, grinding, embodied energy | * 321 | - | 381 | MJ/kg |

## Material processing: climate change

| | | | | |
|---|---|---|---|---|
| Machining, grinding, climate change (CO2-eq) | * 21 | - | 25 | kg/kg |

## Recycling and end of life

| | |
|---|---|
| Recycle | ✗ |
| Functional recycle | ✗ |
| Combust for energy recovery | ✗ |
| Landfill | ✓ |
| Biodegrade | ✗ |

| | |
|---|---|
| Toxicity rating | Non-toxic |
| A renewable resource? | ✖ |

### Environmental notes

Technical ceramics that are used in the pure state, as B4C usually is, are very energy intensive.

## Supporting information

### Design guidelines

Boron carbide starts as a powder, is pressed (with a polymer binder) to the desired shape, then fired at a high temperature, burning off the binder and causing the powder to sinter. Technical ceramics are formed by the following steps.(a) Pressing, isostatic pressing, powder extrusion (for bars and tubes) or powder injection molding (for intricate, high-volume parts).(b) Green-machining in the unfired state, using standard tools.(c) Firing or "sintering" typically at 1550 - 1700 C for 12 to 20 hours; the part shrinks by about 20%.(d) Diamond grinding to achieve tighter tolerance and surface finish: +/- 10 microns is achievable. The cost of a ceramic part is greatly increased if it has to be diamond-ground. Thus design for net-shape sintering, eliminating step (d) is highly desirable. The standard tolerance for as-fired dimensions is +/- 1% or 125 microns, whichever is greater.

### Technical notes

Boron carbide is exceptionally hard, light and wear resistant. Its neutron-absorbing properties make it useful for nuclear shielding.

### Typical uses

Slurry nozzles, light weight body armor, pestles and mortars for hard materials, shot blast nozzles, ceramic tooling dies, ballistic tiles, diamond tool dressing, precision tool parts, thread spinning nozzles.

## Links

ProcessUniverse

Producers

Reference